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CHE 455

HW 7

① $T \approx e^{-2kx}$

$$k = \frac{\sqrt{2m(U-E)}}{\hbar}$$

$$x = 0.5 \text{ nm} = 0.5 \times 10^{-9} \text{ m}$$

$$U = 4.0 \text{ eV}$$

$$m = 9.11 \times 10^{-31} \text{ kg}$$

$$= 4 \times 1.602 \times 10^{-19} \text{ J}$$

$$\hbar = 1.055 \times 10^{-34} \text{ J}\cdot\text{s}$$

$$= 6.408 \times 10^{-19} \text{ J}$$

$$E = 0$$

$$k = \frac{\sqrt{2(9.11 \times 10^{-31})(6.408 \times 10^{-19} - 0)}}{1.055 \times 10^{-34}}$$

$$= \frac{\sqrt{1.167 \times 10^{-48}}}{1.055 \times 10^{-34}} = 1.024 \times 10^{10} \text{ m}^{-1}$$

$$T = e^{-2(1.024 \times 10^{10})(0.5 \times 10^{-9})} = e^{-10.24} = \boxed{3.57 \times 10^{-5}}$$

②

$$x = 0.45 \text{ nm} = 0.45 \times 10^{-9} \text{ m}$$

$$T = e^{-2(1.024 \times 10^{10})(0.45 \times 10^{-9})} = e^{-9.216} = \boxed{9.92 \times 10^{-5}}$$

③

Reducing the gap from 0.5 to 0.45 almost triples the tunneling probability. This sensitivity is what makes STM so precise.